Theory of Elementary Particles: Lecture 1

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Plan and organization of the course, literature
Particle physics
- Experimental studies
- Energy vs. internal structure
- Sources of particles
- A beginning of particle physics: electron scattering

Convention and notation
Kinematics of relativistic particle
- Energy and momentum
- Kinematics of e^-e^+ collision
- S-matrix
- Remarks on Collision in QFT
- Mandelstam Variables
- Laboratory and center of mass frames

Fundamental Particles and Conservation Laws
- Violation of energy-momentum conservation: neutrinos!

Conservation laws
Classification of fundamental particles
- Leptons
- Hadrons
  - Mesons
  - Baryons
- Quarks
  - Color and baryon number conservation

Fundamental particle masses?
Interactions
Higgs particle

Course plan

1. Intuition, kinematics, classification of particles, etc.;
2. S-matrix and cross section formula, QED Feynman rules;
3. Discrete symmetries: C, P, T;
4. Lie groups and representation theory;
5. Quark model, particles spectrum;
6. Parton model;
7. QCD;
8. Goldstone theorem;
9. Gauge theory and spontaneous symmetry breaking – Higgs mechanism;
10. Fermi model;
11. $SU(2) \times U(1)$ symmetry and electroweak interactions;
12. Elements of the standard model;
13. Neutrino oscillations;

Literature

- Ta-Pei Cheng and Ling-Fong Li, *Gauge theory of elementary particle physics*, CLARENDON PRESS-OXFORD 2006.
- M. E. Peskin and D. V. Shroeder, *Introduction to quantum field theory*, Perseus Books 1995.
- J.D. Björken, S.D. Drell, *Relativistic Quantum Mechanics*, McGrawHill Book Company, New York (1965).
- S. Gasiorowicz, *Elementary Particel Physics*, 1966 John Wiley & Sons, Inc., New York.

General idea

Particle physics

- is a branch of physics which studies the nature of particles that are the constituents of what is usually referred to as matter.
- investigates the properties of "the most" basic constituents of matter.

Tools

$\rightarrow$ Micro-world

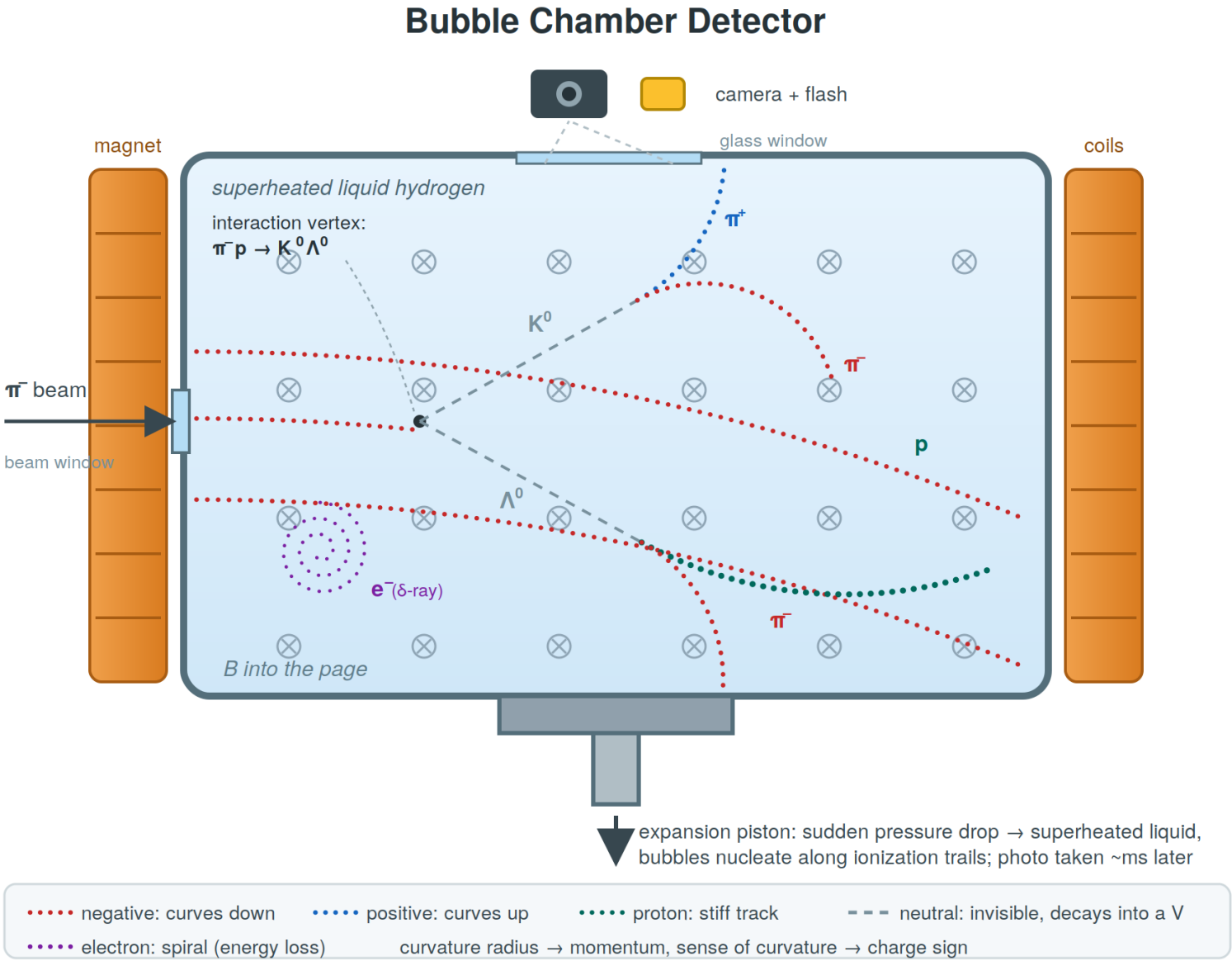
- Quantum mechanics: definite number of particles
- Relativistic mechanics: unbounded number of particles;
- Fundamental symmetries of nature $\rightarrow$ theory of representations of Lie groups
- Poincare group $\rightarrow$ Dirac equation $\rightarrow$ relativistic quantum mechanics $\rightarrow$ concept of antiparticle and creation of particles

$\rightarrow$ **Quantum field theory**

How to investigate the particles properties?

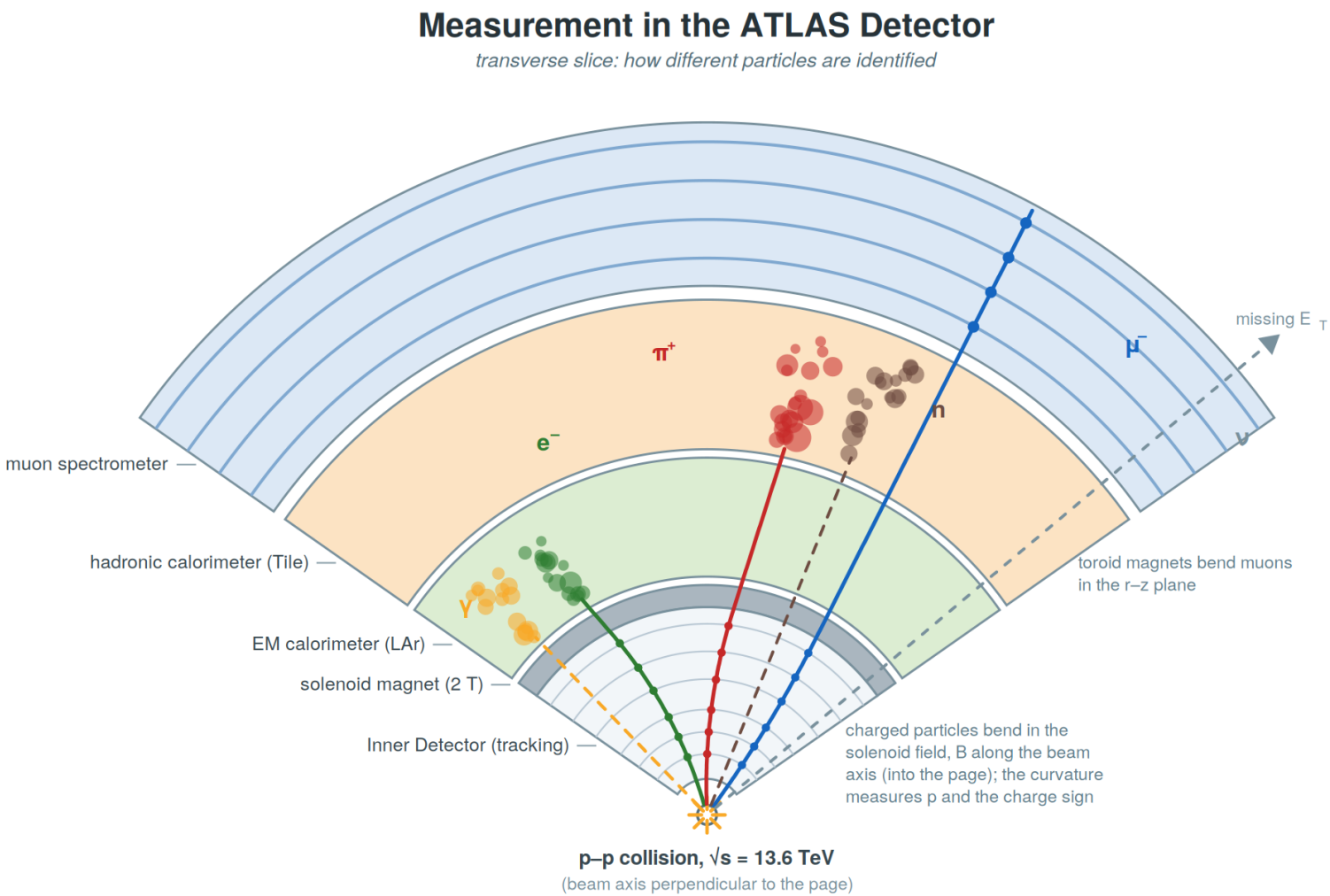
- First: How to study given particle properties?
  - Charged particle: detection of the particle in an external (E-M) field allows to get information about:
  * **charge**, **magnetic moment**, **spin**, **mass**, etc.
  - **Neutral particles**: ???
- Second: How to explore the internal structure of given particle (e.g. proton):
  - Collide two particles and detect the outgoing particle!
  * e.g. study e-proton elastic scattering gives information about the charge distribution inside the proton.
  - Increasing the energy of the incoming particle leads to reduction of wave-length of the probing boson and it improves the "visibility" of the internal structure

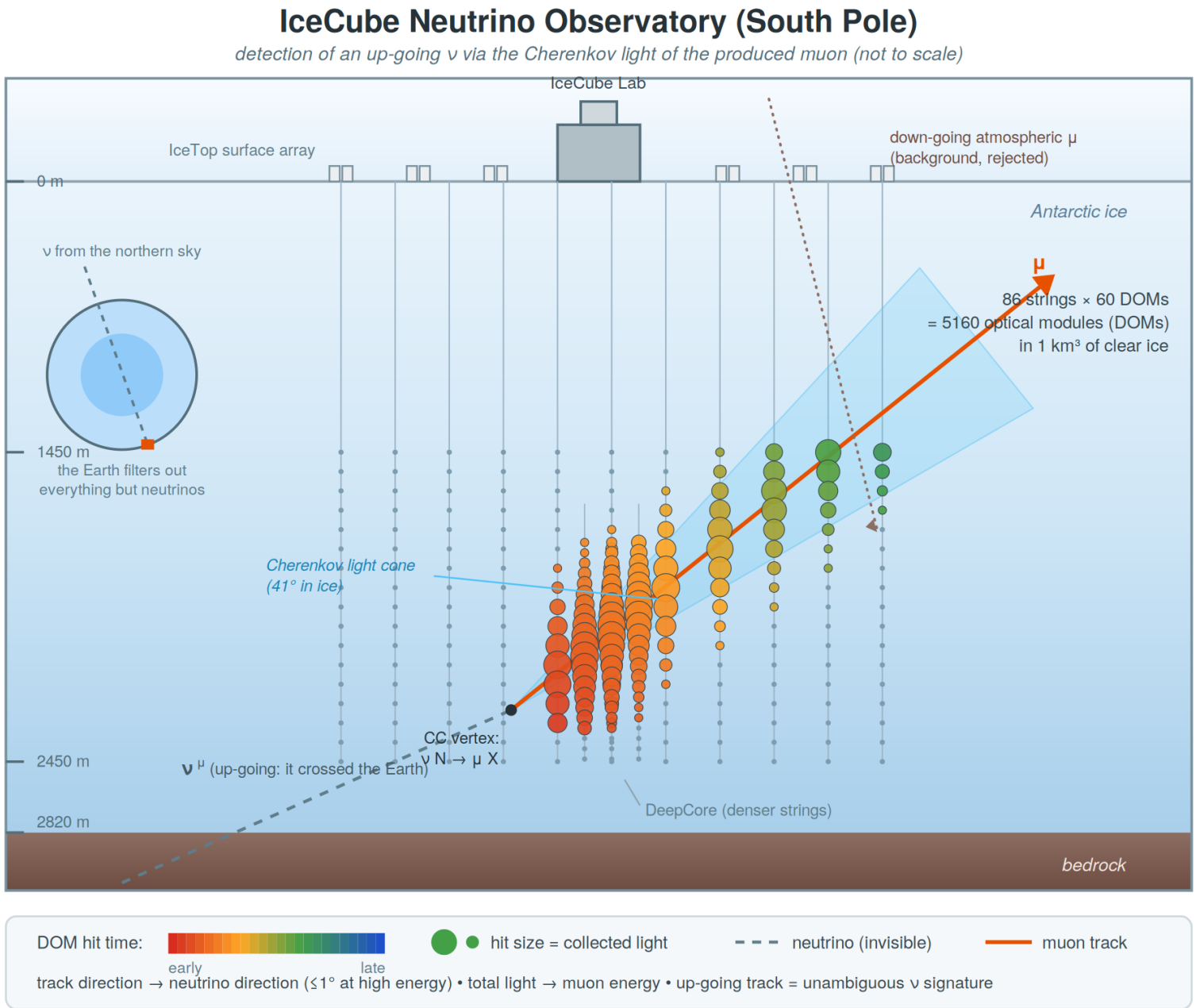
Bubble chamber



Charged particles create an ionization track, around which the liquid vaporizes, forming microscopic bubbles. Bubble density around a track is proportional to a particle's energy loss.

ATLAS @ LHC: Large Hadron Collider (pp)




IceCube Neutrino Observatory (South Pole)
detection of an up-going ν via the Cherenkov light of the produced muon (not to scale)
IceCube Lab
IceTop surface array
down-going atmospheric μ
(background, rejected)
0 m
Antarctic ice
ν from the northern sky
μ
86 strings × 60 DOMs
= 5160 optical modules (DOMs)
in 1 km³ of clear ice
1450 m
the Earth filters out
everything but neutrinos
Cherenkov light cone
(41° in ice)
CC vertex:
ν N → μ X
2450 m
ν μ (up-going: it crossed the Earth)
DeepCore (denser strings)
2820 m
bedrock
DOM hit time:
early
late
hit size = collected light
neutrino (invisible)
muon track
track direction → neutrino direction (≲1° at high energy) • total light → muon energy • up-going track = unambiguous ν signature

Nobel Prize 2026: Francis Halzen



Energy vs. internal structure

- 10^5 fm — Atoms
- several fm — Nuclei
- 1 fm — Nucleons
- point like? — Leptons, Quarks

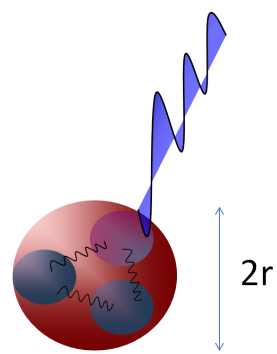
- Wave-Particle duality
- Planck-Einstein relation

$$E = \frac{hc}{\lambda} \tag{1}$$

h Planck's constant, or

$$\lambda = \frac{h}{p} \tag{2}$$

- Let λ denotes the wave-length of the intermediate boson (e.g. γ)
- Typical scale: 1 fm $= 10^{-15}$ m
- λ defines the length scale probed by intermediate boson



Sources of particles

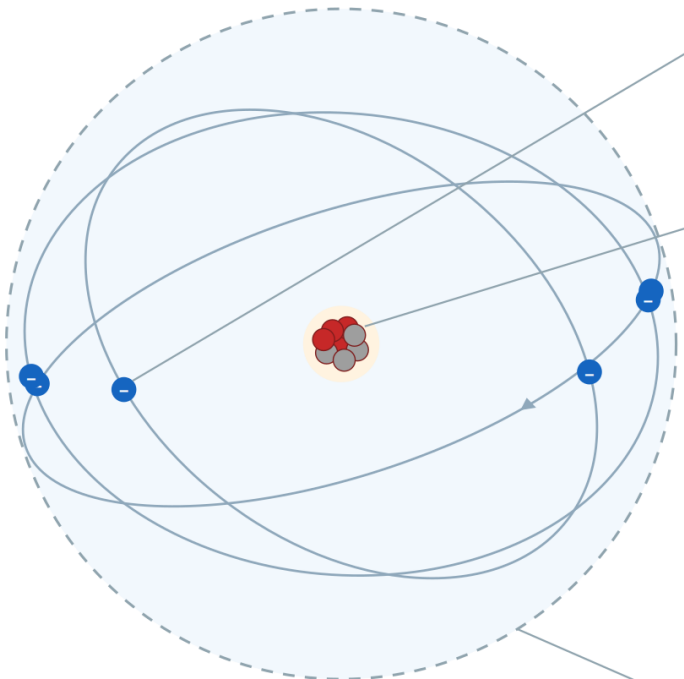
Natural and Artificial

- Astrophysical: Sun, Stars, Atmosphere, Supernovae, etc.
- Natural radioactivity
- Nuclear plants, accelerators, etc.

Rutherford experiment

Rutherford's Model of the Atom (1911)

a tiny, massive, positively charged nucleus — and mostly empty space

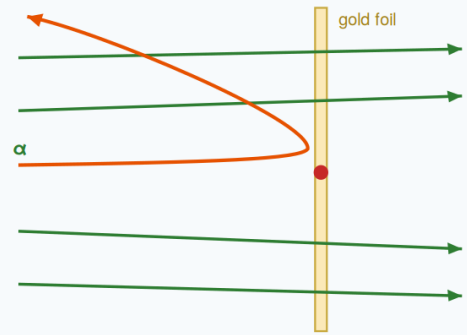


electrons (–e)
orbit the nucleus like planets; the Coulomb force plays the role of gravity

nucleus
charge +Ze, carries ~99.95% of the atomic mass, radius ~ 10^{-14}–10^{-15} m

atom: r ~ 10^{-10} m
the atom is mostly empty space; the nucleus is ~10^{5}× smaller (drawing not to scale!)

The evidence: α scattering off gold



most α pass almost undeflected;
~1 in 8000 bounces back (θ > 90°)

dσ/dΩ ~ 1/sin⁴(θ/2) *(Geiger & Marsden 1909)*

The classical problem

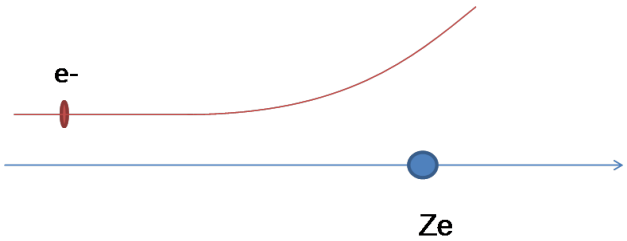


an accelerating electron radiates, loses energy and spirals into the nucleus in ~10^{-11} s — atoms should not be stable!

→ **resolved by Bohr (1913):** quantized orbits

E. Rutherford, "The Scattering of α and β Particles by Matter and the Structure of the Atom", Phil. Mag. 21, 669 (1911)

Scattering in terms of QM and RQM



- Rutherford formula, E. Rutherford, Phil. Mag., **21**, 669 (1911):

$$\frac{d\sigma_{Ruth.}}{d\Omega} = \frac{Z^2 e^4}{16\pi^2 m^2 v_\infty^4} \frac{1}{\sin^4 \frac{\theta}{2}} = \frac{Z^2 \alpha^2}{4E^2} \frac{1}{\sin^4 \frac{\theta}{2}} \tag{3}$$

- Schoedinger equation, solved in CMF (see: Schiff, *Quantum Mechanics*):

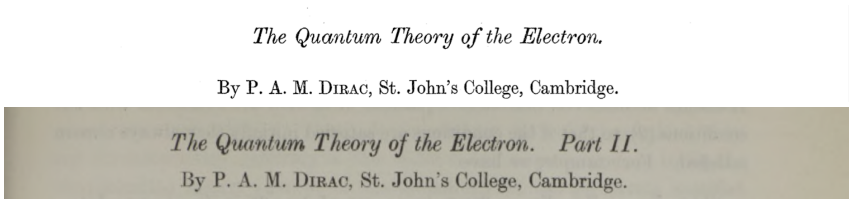
$$\left[-\frac{\hbar^2}{2m} \nabla^2 - \frac{Ze^2}{r} - E \right] u(\mathbf{r}) = 0 \tag{4}$$

- Solution in term of quantum mechanics: particle scatters on the Coulomb potential, number of particles conserved, no new type particle production...
- Suppose that electron is relativistic?

Relativistic theory of electron

- **Special relativity $\rightarrow$ relativistic quantum mechanics**
- Dirac equation: one of the most important achievements in particle physics

$$(i\gamma_\mu \partial^\mu - M)\psi(x^\nu \equiv (t, \mathbf{x})) = 0 \tag{5}$$


The Quantum Theory of the Electron.

By P. A. M. Dirac, St. John's College, Cambridge.

The Quantum Theory of the Electron. Part II.

By P. A. M. Dirac, St. John's College, Cambridge.

- Spin $1/2$ particle (Fermion)
- For every charged particle there exists its antiparticle (positron for electron)!
- Relativistic correction to the energy spectrum in hydrogen atom, not predicted within Klein-Gordon theory and non-relativistic quantum mechanics
- Possible particle production!

Recommended to read:
P. Dirac, Proc. R. Soc. Lond. A**117**, 610 (1928),
and Proc. R. Soc. Lond. A**118**, 351 (1928)

Convention and notation

- Usually we follow the convention and notation of Peskin, Shroeder, *Introduction to quantum field theory*

The metric tensor reads,

$$g_{\mu\nu} = g^{\mu\nu} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix} \tag{6}$$

Four-vector,

$$x^\mu = (x^0, \mathbf{x}) = (ct, \mathbf{x}), \quad x_\mu = (x^0, -\mathbf{x}) \tag{7}$$

$$p_\mu x^\mu \equiv p \cdot x = p^0 x^0 - \mathbf{p} \cdot \mathbf{x} \tag{8}$$

Covariant derivative

$$\partial_\mu = \frac{\partial}{\partial x^\mu} \tag{9}$$

Remark

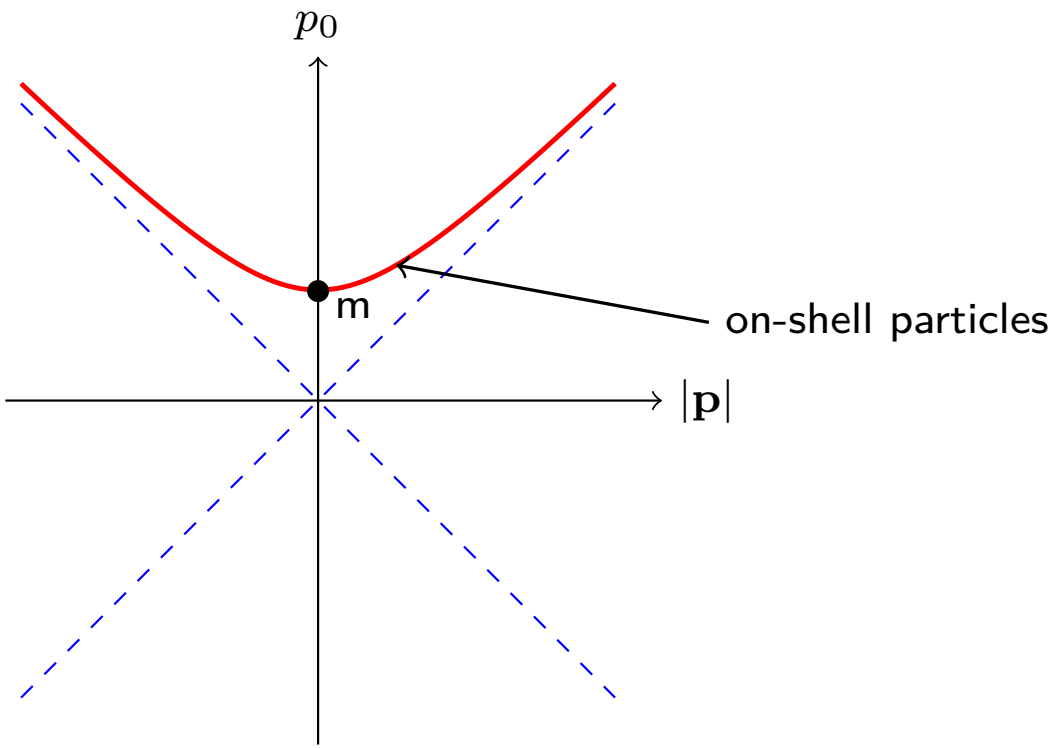
Notice that we work in the unit system with $c = 1$!

Energy and momentum

For the on-shell particle,

$$p^2 = E_p^2 - \mathbf{p}^2 = m^2, \quad E_p \equiv \sqrt{\mathbf{p}^2 + m^2}, \quad \overline{p \cdot x} = E_p x_0 - \mathbf{p} \cdot \mathbf{x}. \tag{10}$$

Velocity of relativistic particle $\mathbf{v} = \mathbf{p}/E$



Physical Particle

- Real freely moving particles are on-shell.
- Virtual particles are off-shell as well as particles propagating in medium...

Energy and momentum

On-shell particle (in contravariant frame):

$$p^{\mu} = (E, \mathbf{p}) \tag{11}$$

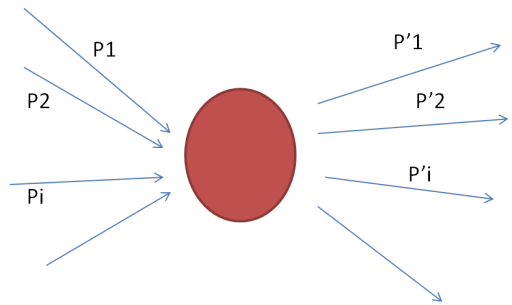
and

$$p_{\mu}p^{\mu} \equiv p^2 \equiv p^{\mu}p^{\nu}g_{\mu\nu} = E^2 - \mathbf{p}^2 \tag{12}$$

Energy and momentum conservation

- **There is no evidence for violation of energy and momentum conservation!**
- Consider the process:

$$P_1(p_1)+P_2(p_2)+...+P_i(p_i)+... \rightarrow P'_1(k_1)+P'_2(k_2)+...+P'_f(k_f)+... \quad (13)$$



- Conservation of energy and momentum:

$$\sum_i p_i^\mu = \sum_f k_f^\mu \quad (14)$$

Notice that it means that one has four equations:

$$\sum_i \sqrt{m_i^2+\mathbf{p}_i^2} = \sum_f \sqrt{m_f^2+\mathbf{k}_f^2}, \quad \sum_i p_i^j = \sum_f k_f^j, \quad j=1,2,3. \quad (15)$$

Kinematics of e^-e^+ collision

Example

Consider

$$e^+(k_A) + e^-(k_B) \to \overline{p}(p_A) + p(p_B) \tag{16}$$

Assume that beams are co-linear (center of mass frame), what is the minimal energy of electron/positron beam in order to produce two protons in final state?.

* We assume that the electron and the positron have the same masses denoted by m_e

** We assume that the proton and the antiproton have the same masses denoted by M

Kinematics of e^-e^+ collision

Notice that

$$p_A + p_B = k_A + k_B \tag{17}$$

Collinear means that $\mathbf{k_A} + \mathbf{k_B} = 0$ hence

$$E_{k_A} = E_{k_B} = \sqrt{m_e^2 + \mathbf{k}_A^2} \equiv E_{e^+/e^-} \tag{18}$$

then

$$(k_A + k_B)^2 = (2E_{e^+/e^-})^2 = 4E_{e^+/e^-}^2 \tag{19}$$

One the other hand

$$4E_{e^+/e^-}^2 = (p_A + p_B)^2 = (2E_{\overline{p}/p})^2 - (\underbrace{\mathbf{p}_A + \mathbf{p}_B}_{=0})^2 = 4E_{\overline{p}/p}^2 = 4(M^2 + \underbrace{\mathbf{p}_A^2}_{\geq 0}) \geq 4M^2 \tag{20}$$

Hence the minimal energy in the beam must exceed $M = 938$ MeV$/c^2 \approx 1$ GeV

Remarks on Collision in QFT

- Let two particles: A and B collide: what is a probability that particles: C, D and E will be produced?

$$Probability(C(k_C), D(k_D), E(k_E)|A(p_A), B(p_B)) \sim |Amplitude|^2 \tag{21}$$

- QFT gives us set of tools for computing the scattering amplitudes, and then cross sections (observables)
- Scattering amplitude

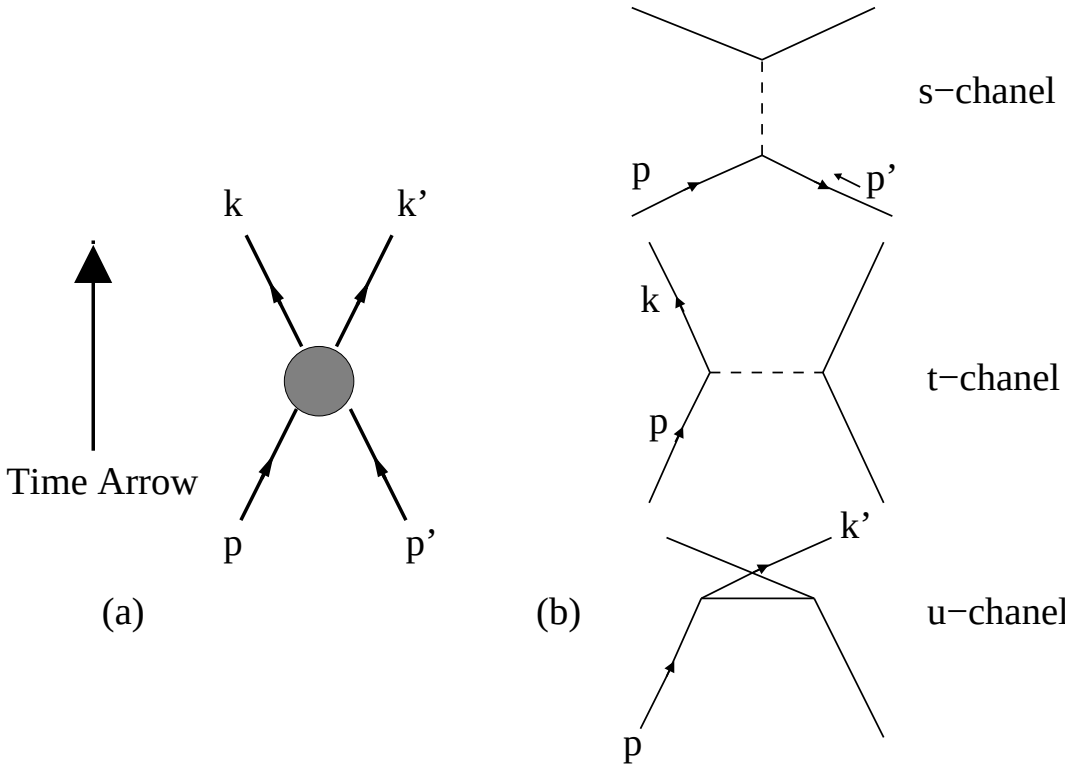
$$Amplitude \sim \langle final \mid S \mid initial \rangle \tag{22}$$

- S-matrix is of main interest
- ... can be expressed in terms of Green's functions etc.
- Lorentz symmetry of QFT
- Transition amplitudes, S-matrices are Lorentz-invariant objects!
- In principle one may express the scalar matrix elements by Lorentz scalars, like Mandelstam variables.

* Kinematics vs. dynamics (described by interactions)

Mandelstam Variables

S. Mandelstam, Phys. Rev. 112, 1344



$$s = (k + k')^2, \quad (23)$$
$$t = (k' - p')^2, \quad (24)$$
$$u = (k' - p)^2. \quad (25)$$

$$i\mathcal{M}_s \propto \frac{1}{s - m_\phi^2},$$
$$i\mathcal{M}_t \propto \frac{1}{t - m_\phi^2},$$
$$i\mathcal{M}_u \propto \frac{1}{u - m_\phi^2},$$

For exercise:

$$s + t + u = \sum_i^4 m_i^2.$$

ϕ is an intermediate particle.

Mandelstam Variables

- t-channel:

$$\nu + n \to p + \mu^-$$

  or

$$e + p \to e + p$$

- s-channel

$$e^- + e^+ \to p + \overline{p}$$

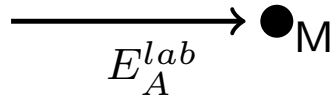
- but in

$$e^- + e^+ \to e^- + e^+$$

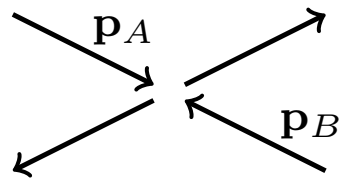
  there are t and s channel contributions

Laboratory and center of mass frames

- One may perform calculations in the most convenient Lorentz frame
- **Laboratory frame:**



  - Here a target is at rest, so $p^{\mu}_{target} = (M, 0, 0, 0)$
  - Situation typical for linear accelerator (Slac)
- **Center of mass frame:**



  - $\mathbf{p}_A + \mathbf{p}_B = 0$
  - Situation typical for e^+e^- collision (LEP) etc.

Relativistic kinematics: the Glashow resonance

S. L. Glashow, Phys. Rev. 118, 316 (1960)

- Resonant formation of the W^- boson in antineutrino–electron scattering:

$$\overline{\nu}_e + e^- \to W^- \to \text{anything} \tag{26}$$

- Target electron at rest (lab frame): $p_e^\mu = (m_e, \mathbf{0})$, neutrino massless: $p_\nu^\mu = (E_\nu, \mathbf{p}_\nu)$, $p_\nu^2 = 0$
- Invariant mass squared of the system (s-channel):

$$s = (p_\nu + p_e)^2 = \underbrace{p_\nu^2}_{=0} + m_e^2 + 2\, p_\nu \cdot p_e = m_e^2 + 2 m_e E_\nu \tag{27}$$

- Resonance condition: $s = M_W^2$, hence

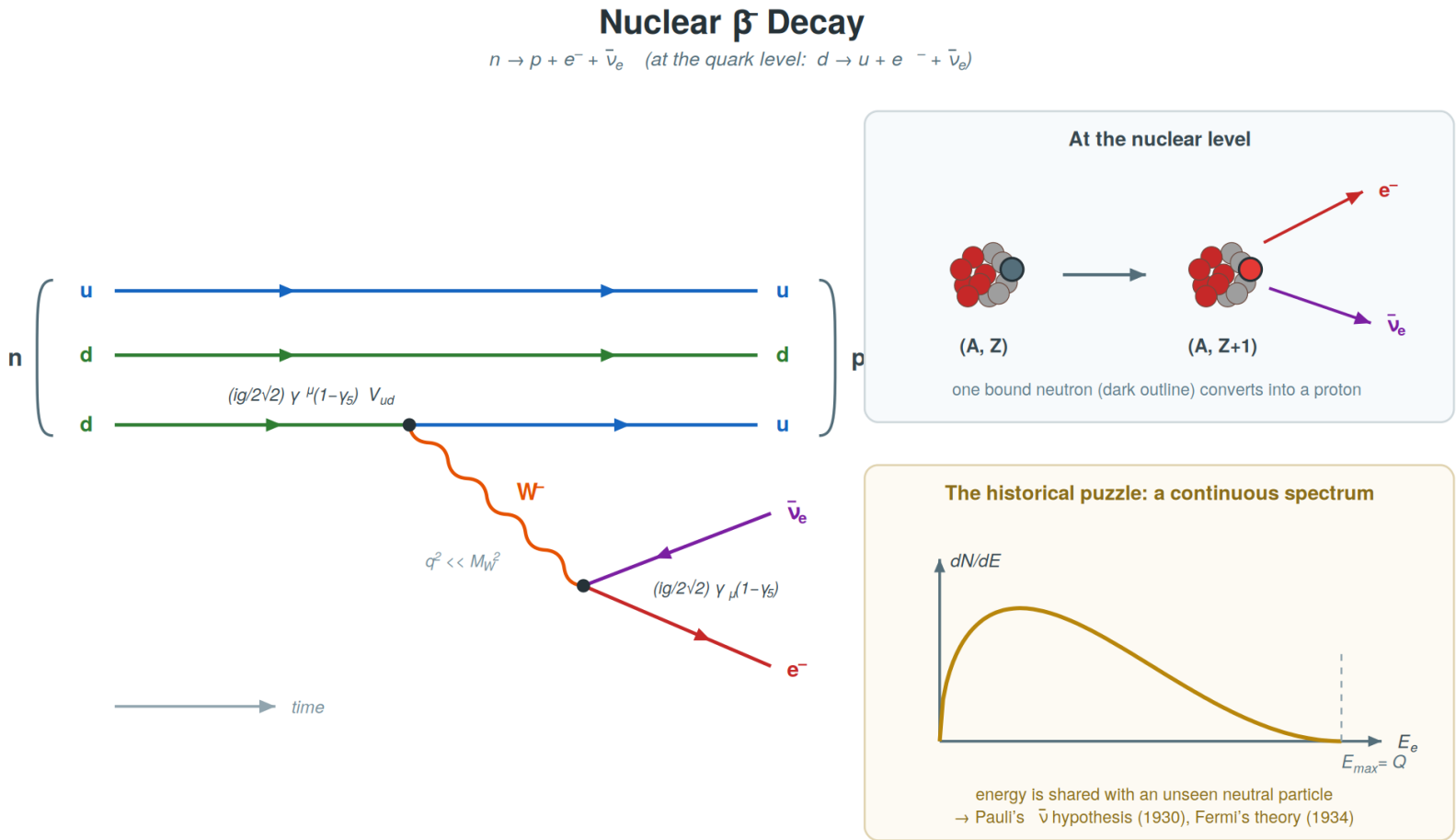
$$E_\nu^{\text{res}} = \frac{M_W^2 - m_e^2}{2 m_e} \simeq \frac{(80.4\ \text{GeV})^2}{2 \times 0.511\ \text{MeV}} \approx 6.3\ \text{PeV} \tag{28}$$

- Cross section enhanced by the Breit–Wigner pole: $\sigma \propto \left[(s - M_W^2)^2 + M_W^2\, \Gamma_W^2 \right]^{-1}$

Remark

$E_\nu^{\text{res}} \sim$ PeV $\Rightarrow$ far beyond accelerators — only astrophysical neutrinos!First candidate event observed by **IceCube**: Nature **591**, 220 (2021), visible energy ≈ 6 PeV.

Violation of energy-momentum conservation: neutrinos!



$$(A, Z) \rightarrow (A, Z + 1) + e \tag{29}$$

Electron should has a fixed kinetic energy approximately equal to release energy

$$E_e = M_{A,Z} - M_{A,Z+1} - m_e \tag{30}$$

but experiment observed continues energy spectrum!

Pauli's proposal



Pauli introduced neutrino (ν):

- energy conservation maintained.
- a new window for physics had been opened!

Pauli:

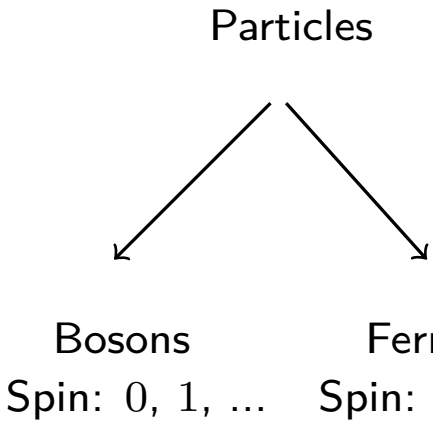
$$(A, Z) \rightarrow (A, Z+1) + e + ''neutron'' \quad (31)$$

$$n \rightarrow p + e^- \Rightarrow n \rightarrow p + e^- + \overline{\nu}_e \quad (32)$$

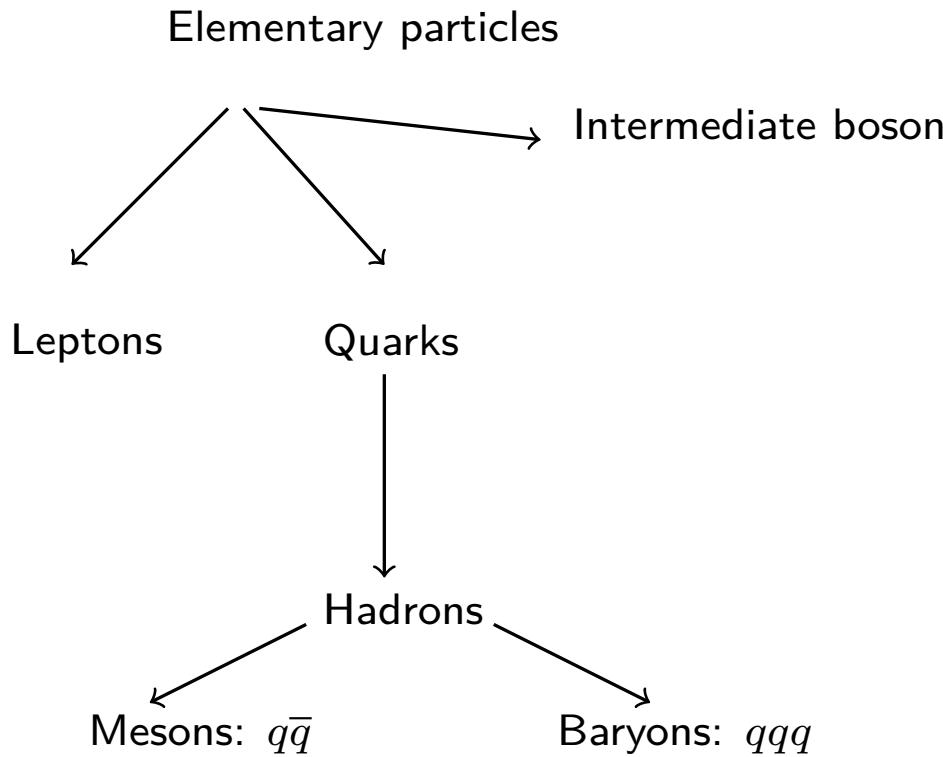
Summary:

- Energy is conserved
- Momentum is conserved
- Charge is conserved

Classification of fundamental particles



- Bosons: γ, $W^{\pm}$, Z^0,..., π, ...
- Fermions: e, μ, ν, q, ...,p, n, ...



Leptons

- Obey the Fermi-Dirac statistics
- Neutrinos: ν_e, ν_μ and ν_τ
  - neutral (charge$= 0$), described either by Dirac or Majorana representations
- Charged leptons: $e^{\pm}$, $\mu^{\pm}$, $\tau^{\pm}$
- Fermions: particle–antiparticle: ν–$\overline{\nu}$, e^-–e^+

$$\begin{pmatrix} \nu_e \\ e^- \end{pmatrix}, \quad \begin{pmatrix} \nu_\mu \\ \mu^- \end{pmatrix} \quad \begin{pmatrix} \nu_\tau \\ \tau^- \end{pmatrix}$$

Three lepton families: Z^0-decay



Leptons

leptons	charge	spin	mass	L_e	L_μ	L_τ
ν_e	0	$1/2$	< 2 eV	1	0	0
e^-	-1	$1/2$	0.51 MeV	1	0	0
$\overline{\nu}_e$	0	$1/2$		-1	0	0
e^+	$+1$	$1/2$		-1	0	0
ν_μ	0	$1/2$	< 170 keV	0	1	0
μ^-	-1	$1/2$	105 MeV	0	1	0
$\overline{\nu}_\mu$	0	$1/2$		0	-1	0
μ^+	$+1$	$1/2$		0	-1	0
ν_τ	0	$1/2$	< 18.2 MeV	0	0	1
τ^-	-1	$1/2$	1777 MeV	0	0	1
$\overline{\nu}_\tau$	0	$1/2$		0	0	-1
τ^+	$+1$	$1/2$		0	0	-1

Note: direct kinematic historical bounds

Lepton number conservation

Addative conservation law

- Lepton number is conserved within each family (approximate symmetry)

$\rightarrow$ ... broken by neutrino oscillations

- For massless neutrinos exact symmetry, according to the Standard Model
- For Dirac neutrinos: Total lepton number conservation $L_e + L_\mu + L_\tau$

Example

$$\pi^+ \rightarrow \mu^+ + \nu_\mu + \rightarrow e^+ + \nu_e + \overline{\nu}_\mu + \nu_\mu \tag{33}$$

	π^+	$\rightarrow$	μ^+	ν_μ	$\rightarrow$	e^+	ν_e	$\overline{\nu}_\mu$	ν_μ
L_e :	0	$\rightarrow$	0	0	$\rightarrow$	-1	1	0	0
L_μ :	0	$\rightarrow$	-1	1	$\rightarrow$	0	0	-1	1
L_τ :	0	$\rightarrow$	0	0	$\rightarrow$	0	0	0	0

Hadrons

- Composite particles not elementary
- Have an internal structure
- Hadrons = Mesons + Baryons

Baryons, B=1

- qqq systems
- Spin:

$$\frac{1}{2} \otimes \frac{1}{2} \otimes \frac{1}{2} = \frac{1}{2} \oplus \frac{1}{2} \oplus \frac{3}{2}$$

- obey Fermi-Dirac statistics

Mesons, B=0

- $q\bar{q}$ pairs
- Spin:

$$\frac{1}{2} \otimes \frac{1}{2} = 0 \oplus 1$$

- obey Bose-Einstein statistics

Mesons

Meson	$I(J^P)$	Mass (MeV/c^2)	Mean Life (s)
$\pi^{\pm}$	$1(0^-)$	139.6	$2.6 \cdot 10^{-8}$
π^0	$1(0^-)$	135.0	$0.8 \cdot 10^{-16}$
η	$0(0^-)$	547.9	$0.8 \cdot 10^{-18}$
$K^{\pm}$	$\frac{1}{2}(0^-)$	493.7	$1.2 \cdot 10^{-8}$
K^0,$\overline{K}^0$	$\frac{1}{2}(0^-)$	497.7	
K_S^0			$0.9 \cdot 10^{-10}$
K_L^0			$5.2 \cdot 10^{-8}$

K^0,$\overline{K}^0$ are the massive states which are given by linear combination of K_L^0 and K_S^0 states

- P parity
- I isospin
- J spin

Baryons

Baryon	$I(J^P)$	Mass (MeV/c^2)	Mean Life (s)
p	$\frac{1}{2}\left(\frac{1}{2}^+\right)$	938.3	$> 10^{31}$ yrs
n	$\frac{1}{2}\left(\frac{1}{2}^+\right)$	939.6	879
Λ^0	$0\left(\frac{1}{2}^+\right)$	1115.6	$2.6 \cdot 10^{-10}$
Σ^+	$1\left(\frac{1}{2}^+\right)$	1189.4	$0.8 \cdot 10^{-10}$
Σ^0	$1\left(\frac{1}{2}^+\right)$	1192.5	$7.0 \cdot 10^{-20}$
Σ^-	$1\left(\frac{1}{2}^+\right)$	1197.4	$1.5 \cdot 10^{-10}$
Ξ^0	$\frac{1}{2}\left(\frac{1}{2}^+\right)$	1314.9	$2.9 \cdot 10^{-10}$
Ξ^-	$\frac{1}{2}\left(\frac{1}{2}^+\right)$	1321.3	$1.6 \cdot 10^{-10}$

Table: The lowest laying baryons.

Quarks

quarks	charge	spin	mass	constituted mass (GeV/c^2)
u	$2/3$	$1/2$	2.3 MeV/c^2	0.33
d	$-1/3$	$1/2$	4.8 MeV/c^2	0.33
c	$2/3$	$1/2$	1.27 GeV/c^2	1.5
s	$-1/3$	$1/2$	95 MeV/c^2	0.5
t	$2/3$	$1/2$	172.7 GeV/c^2	180
b	$-1/3$	$1/2$	4.18 GeV/c^2	4.5

Three quark families:

$$\begin{pmatrix} u \\ d \end{pmatrix}, \quad \begin{pmatrix} c \\ s \end{pmatrix}, \quad \begin{pmatrix} t \\ b \end{pmatrix} \tag{34}$$

Color and baryon number conservation

Quarks:

- Fermions, spin-$1/2$
- every quark state is characterized by color quantum number: red,yellow,blue, it plays an analogical role as an electric charge in QED
- quarks can not exist in isolation, they have to form clusters with the total charge $\pm ne$, where n is integer.
- hadrons are colourless (singlets due to colour group) antiquark has opposite colour to corresponding quark
  - $q\overline{q}$ =colour+(−)colour = white
  - qqq=redyellowblue = white

	Baryon Number
q	$\frac{1}{3}$
$\overline{q}$	$-\frac{1}{3}$
meson	0
baryon	1
lepton	0

- Baryon number is conserved in all observed processes.

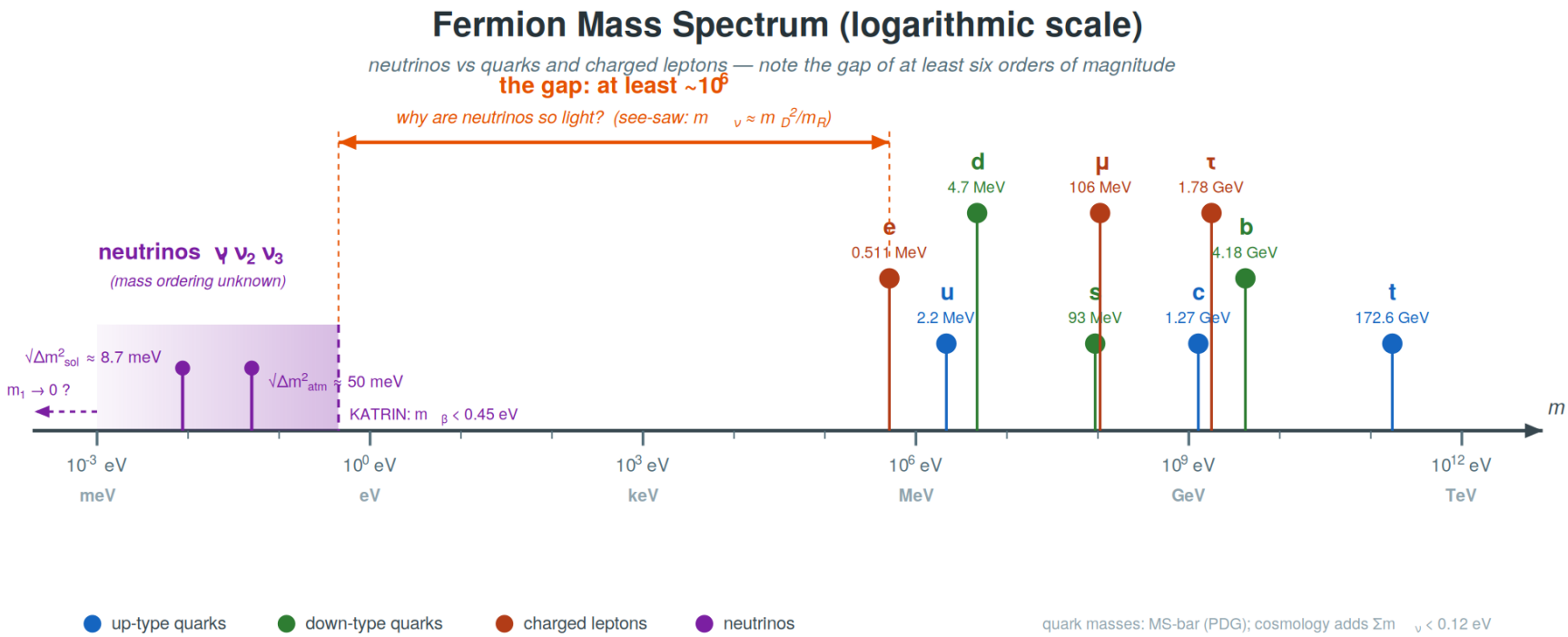
Example:

$$\begin{array}{rcccccc} \nu + p & \rightarrow & \Delta^+ + e^- & \rightarrow \pi^+ + p + e^- \\ B: 0 + 1 & \rightarrow & 1 \quad + 0 & \rightarrow 0 \quad + 1 + 0 \end{array}$$

Conservation laws

A summary

- Energy is conserved
- Momentum is conserved
- Charge is conserved
- Lepton number conservation
- Baryon number conservation
- Color is conserved



What is the mechanism for neutrino mass generation?!

What fundamental forces we know? What can you tell about them?

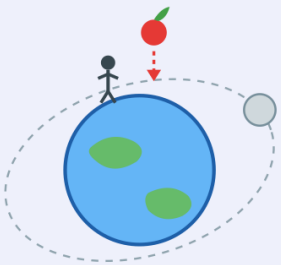
Four Fundamental Interactions

The Four Fundamental Interactions

what each of them is responsible for in everyday life

GRAVITY

always attractive — it rules everything big



- keeps us (and the air) on the Earth
- the falling apple, ocean tides, the orbits of the Moon and planets
- the weakest of all — yet it rules the cosmos, because it never cancels

range: ∞ | strength: $\sim 10^{-39}$

ELECTROMAGNETISM

all of everyday life apart from gravity

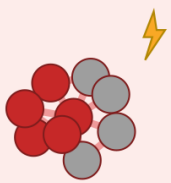


- light, electricity, magnets, Wi-Fi
- chemistry: binds atoms into molecules — that is why a chair stops your hand
- friction, elasticity, colours, all of electronics and biology

range: ∞ | strength: $\sim 10^{-2}$

STRONG INTERACTION

the nuclear "glue" — strongest, but very short-ranged



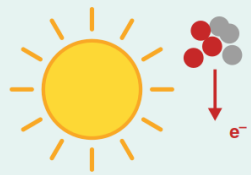
the gluon "glue"

- glues quarks into protons and neutrons, and protons and neutrons into nuclei
- without it — no chemical elements
- nuclear energy: power plants, bombs and the cores of stars

range: $\sim 10^{-15}$ m | strength: 1 (reference)

WEAK INTERACTION

the only one that can change one particle into another



the Sun shines thanks to it

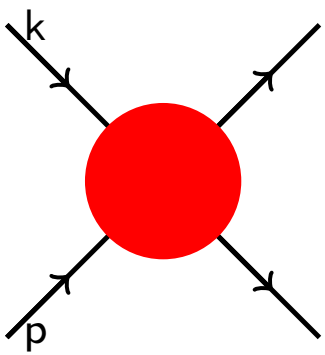
- radioactive (β) decays
- ignites the Sun: turns a proton into a neutron in hydrogen fusion
- radiocarbon C-14 dating (archaeologists!), nuclear medicine

range: $\sim 10^{-18}$ m | strength: $\sim 10^{-6}$

force carriers: graviton (hypothetical) · photon · gluons · W and Z bosons

Fundamental Forces

- Electromagnetic interactions (E-M): every day phenomenon
- Weak e.g. β-decay
- Strong interactions: keep nucleons, nuclei in one peace
- Gravity



$$\sim \frac{1}{(k+p)^2 - M_{eff}^2} \sim \langle f|S|i\rangle$$

A pole at physical mass M_{eff}^2 of the S-matrix can be interpreted as the exchange of the off-shell particle of mass M_{eff} between interacting particles.

Interactions and intermediate Bosons

Interactions and QFT

Interactions between particles are induced by an exchange of energy quanta which have all attributes of ordinary matter particles.

Boson		Force	Symmetry	Spin	Mass (GeV/c^2)	Q
A_μ	photon	E-M	$U(1)$	1	0	0
$W_\mu^\pm$		Weak	$SU(2) \times U(1)$	1	80	± 1
Z_μ^0		Weak	$SU(2) \times U(1)$	1	91	0
G_μ^i $i = 1, ..., 8$	gluon	Strong	$SU(3)$	1	0	0

Interactions

Interactions	Effective coupling	Boson	M (GeV/c^2)	Range (cm)	Time (s)
Gravity	10^{-39}	graviton	0	∞	–
E-M	$1/137$	photon	0	∞	10^{-20}
Weak	10^{-5}	$W^{\pm}, Z^0$	80, 90	10^{-16}	10^{-10}
Strong	1	gluons	0	$< 10^{-13}$	10^{-23}

If two particles exchange virtual particle of mass m then $\Delta E = mc^2$ is at the minimal energy transferred, then

$$\Delta t = \frac{\hbar}{\Delta E} = \frac{\hbar}{mc^2}$$

Assuming that the particles move with velocity of light, then the effective range reads

$$c\Delta t = \frac{\hbar}{mc}$$

Higgs Particle

- Discovered at CERN in 2012
- $m_H \approx 125$ GeV
- neutral particle
- spin 0
- esponsible for the masses of the elementary fermions and the $W^{\pm}, Z$ bosons in the Standard Model; most of the mass of hadrons originates dynamically from QCD.

https://twiki.cern.ch/twiki/pub/AtlasPublic/HiggsPublicResults/
/4l-FixedScale-NoMuProf2.gif